

Which of the following is true about the structure and reactivity of steroids?

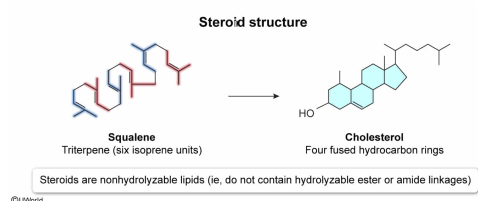
- I. Steroids contain only three fused hydrocarbon rings
- II. Steroids are made up of two isoprene units
- III. Steroids are nonhydrolyzable lipids

- ✓ ☒ A. III only [56%]
☐ B. I and III only [9%]
☐ C. II and III only [29%]
☐ D. I, II, and III [4%]

Correct

55% answered correctly

Explanation:



Terpenes are precursors to steroids, which are lipids involved in signaling pathways. Individual terpenes are made up of two or more 5-carbon groups known as isoprenes and are **classified** based on the number of isoprene units present.

Terpenes can be joined together in one of **three ways**. Those containing two isoprenes are known as monoterpenes, and molecules made up of two monoterpenes (four isoprenes) are called diterpenes. **Steroids** are made up of **triterpenes** (three monoterpenes, or **six isoprene units**), which are synthesized by the **cyclization of the triterpene squalene**, forming cholesterol. Cholesterol itself contains **four fused hydrocarbon rings** (three 6-membered and one 5-membered rings).

Lipids can be classified based on whether they contain any hydrolyzable linkages (ie, an **ester** or **amide**). Lipids containing an ester or amide (eg, fatty acids, phospholipids) are classified as **hydrolyzable lipids** whereas lipids that do not have an ester or amide (eg, steroids) are known as **nonhydrolyzable lipids**. Steroids do not contain any hydrolyzable linkages, and as such they are nonhydrolyzable lipids (**Number III**).

(Number I) Steroids contain *four* rather than only three fused hydrocarbon rings.

(Number II) Steroids are made from triterpenes, which in total contain *six* (not two) isoprene units.

Educational objective:

Steroids are lipids made from triterpenes, which contain six isoprene units that cyclize to form four fused hydrocarbon rings. Steroids are classified as nonhydrolyzable lipids because they do not contain any hydrolyzable ester or amide linkages.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions